



Technische Hochschule  
Ingolstadt

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## Bachelor Thesis

Faculty of computer science

**Development of a Frontend Application for Real-Time and  
24-Hour Monitoring and Recording of Electrocardiographic Data  
with Communication via Bluetooth Low Energy**

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## AFFIDAVIT

I hereby declare that I have written this thesis independently, have not used any sources or aids other than those stated, have not submitted it elsewhere for examination purposes and have expressly identified all quotations from the sources used, either verbatim or in spirit. I have marked verbatim and indirect quotations as such.

Ingolstadt, \_\_\_\_\_

Firstname Lastname

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## ABSTRACT

The human heart is the central organ of the circulatory system and plays a vital role in supplying the body with oxygen and nutrients. Rhythmic contractions pump blood through the body, laying the foundation for all vital functions. An essential part of medical diagnostics combined with heart function and health is the electrocardiogram (ECG). The ECG measures the electrical activity of the heart and provides valuable information about the heart rhythm as well as possible irregularities or diseases. As technology has advanced, the way in which ECG data can be collected and monitored has also evolved significantly. Mobile health monitoring in particular now makes it possible to record ECG data not only at certain points of time, but also in real time and over longer periods of time. Wireless transmission via Bluetooth Low Energy (BLE) enables reliable and energy-efficient transfer of data to mobile devices. This opens up new possibilities for applications offering real-time and long term monitoring enabling users access to relevant information and graphical processing and alarm functions. This thesis explains the conception and implementation of an Android frontend application, which realises continuous and mobile ECG monitoring. The main programming language used for this project, is Kotlin with Jetpack Compose as the Framework. Moreover this thesis is particularly focused on implementing the communication and data transmission via BLE as well as realising a 24-hour-mode, which might be used for 24-hour data recognition. It also provides a broad insight into the given requirements, design principles, and technical challenges which occurred during the implementation. Furthermore this thesis offers a glimpse into the potential for modern health surveillance.

## ACRONYMS

**API** Application Programming Interface.

**ATT** Attribute Protocol.

**BLE** Bluetooth Low Energy.

**BPM** Beats per Minute.

.

**ECG** Electrocardiography.

**EEG** Electroencephalography.

**EMG** Electromyography.

**GAP** Generic Acces Profile.

**GATT** Generic Attribute Profile.

**GFSK** Gaussian Frequency Shift Keying.

**GHz** Gigahertz.

**HCI** Host Controller Interface.

**Hz** Hertz.

**IDE** Integrated Development Environment.

**ISM** Industrial, Scientific and Medical.

**L2CAP** Logical Link and Adaption Layer Protocol.

**LL** Link Layer.

**MAC** medium acces control.

**Mbps** Megabits per Second.

**MTU** Maximum Transmission Unit.

**OS** Operating system.

.

**SDK** Software Development Kit.

**SMP** Security Manager Protocol.

**UI** User Interface.

**URI** Uniform Resource Identifier.

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## 1. INTRODUCTION

This chapter will give a small introduction on what the thesis and the implemented application is about. This covers an explanation for the application's motivation and the environment the application will be used in.

### 1.1. Motivation

The continuous monitoring of electrocardiographic (ECG) signals plays a crucial role in modern medicine, especially in the early detection and diagnosis of heart disease. While traditional ECG devices are often stationary or bulky, mobile technologies enable flexible and location-independent measurement. This opens up new possibilities for preventive health monitoring and telemedical care of patients. With the increasing use of wearable sensors and Bluetooth Low Energy (BLE) technology, new challenges arise in real-time data processing and visualization. A reliable and efficient software solution is required to capture, analyze and clearly display large amounts of ECG data in real time. The aim of this work is to develop a modern and high-performance front-end application that enables precise and intuitive visualization of ECG data. This work thus makes an important contribution to digital healthcare and demonstrates the potential of mobile technologies for cardiology diagnostics.

Furthermore the goal is to create a new version of an Android App once developed by fellow students from the faculty of Electrical and Information Technology. It's core functionality is to receive and display ECG data in real time, via BLE. With the completion of this work, the new version will be able to receive and display ECG data in real time, via BLE. In addition it will be capable of an 24-hour mode which records ECG data for up to 24 hours, so the collected data can be analyzed later.

### 1.2. Scope and Framework

This paper only deals with the app development process of the new version. Therefore, any implementation of the previous work done by the students is not going to be discussed. The hardware implementation is also not an element of this work. Generally speaking, this paper covers all topics that are essential for the development of the new app version. Specifically, technology stack of choice, IDE choice decision, communication with hardware via Bluetooth low energy, real-time data monitoring, implementing the 24 hour mode. In addition, the outlook explains procedures for the implementation of future features that could not be realized within the limits of this paper.

### 1.3. Procedure

The second chapter, *Technical Fundamentals*, explores the Android architecture, with a particular focus on the Java Framework API. Giving valuable insights into the core functionality of the operating system aswell as an glimpse into the BLE-Stack, laying the groundwork for a better understanding of the implementation details presented later. Chapter three delves into the functional requirements, systematically breaking them down into distinct sub-requirements. These requirements are revisited in chapter four, where they are translated into actual code, accompanied by relevant code snippets to illustrate their implementation.

## 2. TECHNICAL FUNDAMENTALS

### 2.1. Basics of Android architecture

To program a platform-specific application, it can be important to understand the basic architecture of the platform, especially when dealing with tasks regarding memory or Bluetooth. This section will give a brief insight into the structure of the Android construction. Tasks discussed later in this thesis may reference this chapter to convey a better understanding.

The core of the Android Platform is a Linux based kernel. On top of Android's Kernel are four more stack layers, such as the Hardware Abstraction Layer, Native Libraries/Android Runtime, Java API Framework and the uppermost layer, the System App layer. (also see: Figure 1). Going into detail for each stack layer is not necessary for this work, consequently only essential layers such as the Java API Framework will be considered in more depth. A short introduction into each layer is given by table 1 and figure 1 .

**Table 1:** Layers explanation

| Layer                                | Functionality                                                                                                                                                                                                                    |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System App's (Application Layer)     | This layer is a collection of preinstalled applications. Apps contained in this collection, have the same status at system level as third-party apps(system settings is the only exception). Giving the user freedom for design. |
| Java API Framework (Framework Layer) | Enables acces to all features of the Android OS, such as resource manager and view system.                                                                                                                                       |
| Native Libraries                     | System components such as ART and HAL are written in C/C++, those functionalities can be used via API's                                                                                                                          |
| Android Runtime (ART)                | Every running application is an instance of ART. ART operates on the same layer as Native Libraries                                                                                                                              |
| Hardware Abstraction Layer (HAL)     | Provides interfaces to the Java API Framework, hence addressing hardware components like Camera or Bluetooth module is possible.                                                                                                 |
| Linux Kernel                         | The foundation of the OS. Administrates system resources, and communicates with hardware.                                                                                                                                        |

<sup>1</sup> A layered architecture such as the Android OS, ensures modularity, maintainability and security. Each layer is designed with a specific responsibility, and communicates primarily with adjacent layers. Applications interact with the Framework layer via the Android SDK (bridge between Application Layer and Framework Layer ), which provides access to core

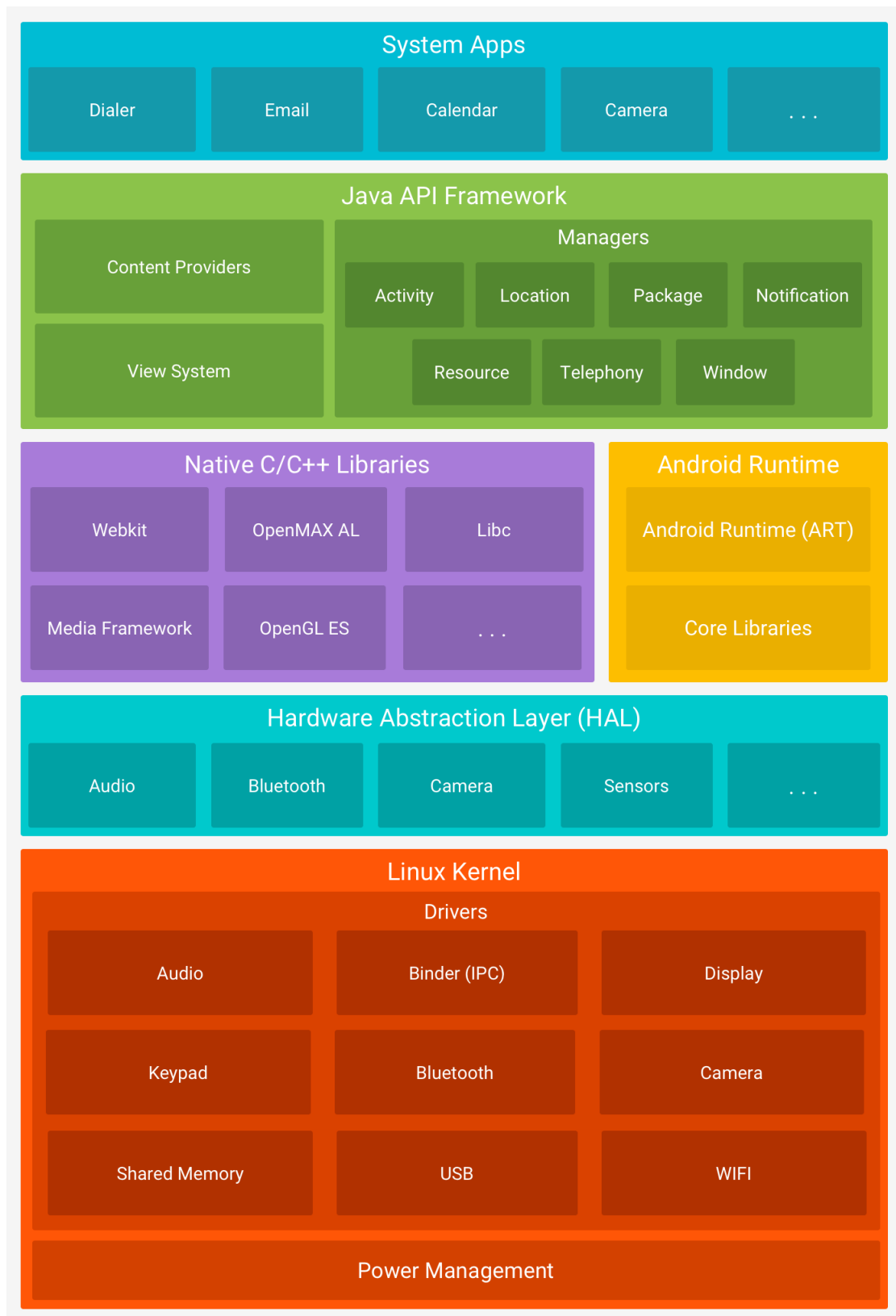
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<sup>1</sup>Portions of this page are modifications based on work created and shared by the Android Open Source Project and used according to terms described in the Creative Commons 2.5 Attribution License.

system services. The Framework, in turn, relies on the Native Libraries and Runtime layers for functionalities like graphics rendering, media playback, and app execution. Hardware communication is abstracted through the Hardware Abstraction Layer (HAL), which interfaces with the Linux Kernel. This structured separation of concerns enhances the system's flexibility, allowing independent evolution and optimization of each layer. [Android \[2025a\]](#)<sup>2</sup>

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<sup>2</sup>Portions of this page are modifications based on work created and shared by the Android Open Source Project and used according to terms described in the Creative Commons 2.5 Attribution License.



**Figure 1:** Android Platform architecture (Source: <https://developer.android.com/guide/platform>)

### 2.1.1. Java Framework API (Framework Layer)

#### Description

As already mentioned in 2.1, the Android Java Framework API is a collection of libraries and tools provided by Android to simplify the development of mobile applications. Built on top of Java, it enables developers to create apps with pre-built components, utilities, and services that interact with the Android operating system. The framework abstracts the complexity of low-level operations and provides high-level APIs for features such as UI design, data storage, networking, and hardware access.

#### Project example

To provide an impression, Figure 2 shows a code excerpt in which the Java API was implicitly used by androidx. To keep this chapter manageable, it can be simplified said that androidx is a superset of the Java API Framework. [Android \[2025b\]](#)

```

1      import androidx.compose.foundation.Image
2      import androidx.compose.foundation.layout.padding
3      import androidx.compose.foundation.layout.size
4      import androidx.compose.material3.Button
5      import androidx.compose.material3.ButtonDefaults
6      import androidx.compose.runtime.Composable
7      import androidx.compose.runtime.getValue
8      import androidx.compose.runtime.mutableStateOf
9      import androidx.compose.runtime.remember
10     import androidx.compose.runtime.setValue
11     import androidx.compose.ui.Modifier
12     import androidx.compose.ui.graphics.Color
13     import androidx.compose.ui.graphics.painter.Painter
14     import androidx.compose.ui.platform.LocalConfiguration
15     import androidx.compose.ui.res.painterResource
16     import androidx.compose.ui.unit.dp
17
18     import com.example.ble.R
19     import com.example.ble.ui.view.ble.BLEViewModel
20     import com.example.ui.view.graph.GraphEcgWidget
21     import com.example.ble.ui.view.graph.BPMWidget
22
23
24     class MainView {
25         @Composable
26         fun BluetoothButton(btButtons: @Composable () -> Unit){
27             val image: Painter = painterResource(id = R.
                drawable.bluetoothicon)

```



```

28         var toggleButtons by remember { mutableStateOf(
29             false) }
30         Button(onClick={toggleButtons = !toggleButtons}){
31             Image(
32                 painter = image,
33                 contentDescription = "Button Image",
34                 modifier = Modifier.size(24.dp)
35             )
36             if(toggleButtons){
37                 btButtons()
38             }
39         }

```

**Code 1:** Android Java API example with androidx

Code 1 provides a glimpse of the capabilities of the androidx library. Showing how to draw an image with the Painter class and utilizing the resource manager mentioned in table 1.

To begin its elementary to import the Painter class as well as the painterResource class (line 13/15). Then at line 26 a Painter object is created, as well as defined by the painterResource function. The painterResource function gets an id as an argument, to query the file from the res/drawable folder. The res folder contains all additional app resources (files, images, etc.). Finally the Image can be drawn, by passing the Painter object to an Image function.

## 2.2. BLE

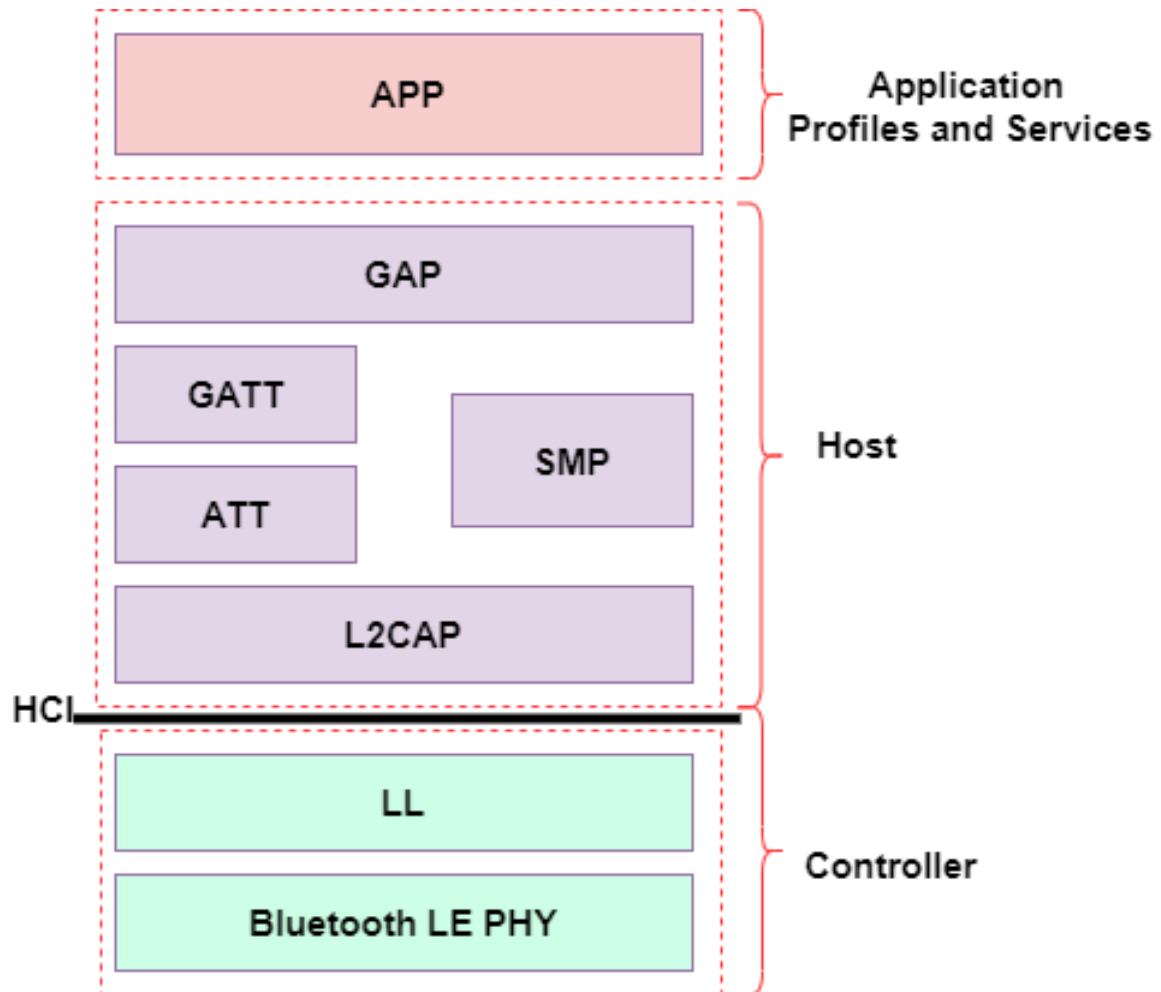
Thus the hardware is a wearable, it is important to manage energy consumption. BLE is designed to minimize energy usage, which is crucial for wearable devices that require extended battery life. A study demonstrated that low-power wearable sensors equipped with BLE could effectively acquire electrophysiological signals like ECG, EMG, and EEG without significant battery drain. [Li et al. \[2018\]](#)

### 2.2.1. BLE Stack

The BLE protocol stack consists of three main layers: the Controller, the Host, and the Application. The Controller encompasses the Physical Layer and the Link Layer, while the Host manages higher-level functionalities such as the Logical Link Control and Adaptation Protocol (L2CAP), the Attribute Protocol (ATT), the Generic Attribute Profile

(GATT), the Security Manager (SM), and the Generic Access Profile (GAP). Communication between the Host and the Controller follows the standardized Host Controller Interface (HCI). At the top of the stack, the Application layer enables program execution.

Dian et al. [2018]



**Figure 2:** BLE protocol stack (Source: <https://de.mathworks.com/help/bluetooth/gs/bluetooth-protocol-stack.html>)

### Application Layer

The application layer serves as the direct user interface, defining profiles that enable interoperability between various applications.

### Host Layer

Includes the host-sided HCI as well as L2CAP, ATT, GATT, SMP and GAP.

### Generic Access Profile (GAP)

Manages the connection setup as well as security, interfaces directly with profiles and services of the Application Layer.

### Security Manager Protocol (SMP)

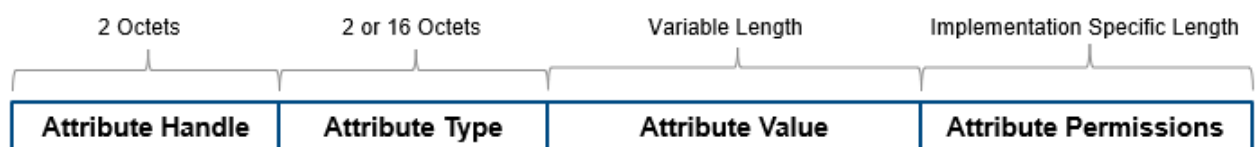
Provides multiple security algorithms for encrypting and decrypting data packets, making the connection secure.

### Generic Attribute Profile (GATT)

Defines services and characteristics provided by the GATT Server. The GATT Server in this case is the microcontroller, which holds the definitions of services and characteristics. These can be requested by the GATT Client (in this case the application). Services organize data into logical entities and contain specific data units known as characteristics. Each service can include one or more characteristics and is uniquely identified by a numeric UUID. This UUID can be either 16-bit for officially adopted BLE services or 128-bit for custom services. Whereas the Characteristic is the lowest level concept in GATT transactions, which defines data. Services as well as Characteristics are main interaction points to work with when working with the BLE periphery, thus it is important to understand these terminologies.

### Attribute Protocol (ATT)

The ATT enables the exchange of attributes (characteristics). Attributes are composed of four components, the type (128-bit UUID), an handle (16-bit identifier) to reference attributes, rights of the attribute (reading, writing), and the value of the attribute. The following figure gives a good overview of the structure of the attributes.



**Figure 3:** Attribute Structure (Source: <https://de.mathworks.com/help/bluetooth/gs/bluetooth-protocol-stack.html>)

### Logical Link Control and Adaptation Layer Protocol (L2CAP)

Works as an interface between the HCI and the upper modules of the Host layer, by

multiplexing data packets to the corresponding protocols. Also fragments and reassembles large packets.

#### **Host-Side Host-Controller-Interface (HCI)**

The HCI is the interface between the host and the controller. The HCI establishes commands and events for packet data transmission and reception. When sending data, it processes raw information into packets, enabling communication between the host and the controller.

#### **Controller-Side Host-Controller-Interface (HCI)**

As mentioned in the last paragraph, the HCI is the interface between the host and the controller. However, analogous to the host side, the data received from the controller is extracted and forwarded to the host.

#### **Controller Layer**

Includes the Bluetooth LE PHY, LL, and controller-side HCI

#### **Link Layer (LL)**

The LL functions similarly to the medium access control (MAC) layer in the OSI model. In Bluetooth, it directly interacts with the Bluetooth LE PHY and controls the radio's link state, determining whether a device operates as a Central, Peripheral, Advertiser, or Scanner.

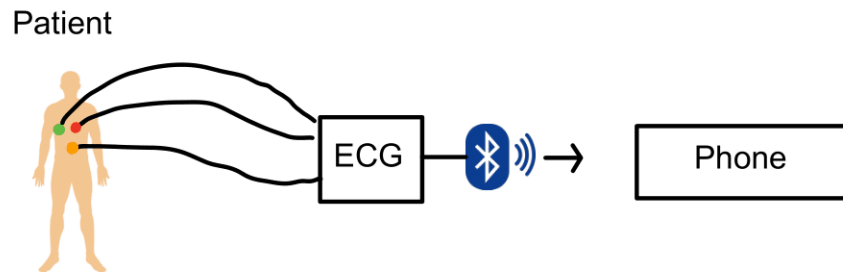
#### **Bluetooth LE PHY**

The physical layer (BLE PHY) is a 1 Mbps GFSK-based radio with adaptive frequency hopping, operating in the unlicensed 2.4 GHz ISM (industrial, scientific, and medical) band. [MathWorks \[2025\]](#)

### **3. FUNCTIONAL REQUIREMENTS**

Requirement engineering is essential at the beginning of every project. Formulating and documenting precisely defined requirements is a cornerstone in the course of realization. In this way, mundane mistakes can be avoided and the vision of the final product is clear. Thus this chapter will explain all functional requirements of the finished product.

### 3.1. ECG-Data



**Figure 4:** Informationtransfer

The application receives integer values from the microcontroller, which represent the heartactivity of the person connected to the hardware. These values must be mapped onto the screen to provide a clear and accurate visualization of the heart's activity. A key challenge in this process is ensuring that the visualization adapts to different screen sizes. Since the application may run on devices with varying display dimensions and resolutions, a static mapping approach is insufficient. Instead, a dynamic and scalable algorithm is required to transform the received data into a graphical representation that maintains consistent proportions and readability across all screen sizes. To achieve this, the application must implement a flexible coordinate transformation. The incoming data points need to be scaled appropriately based on the available screen width and height while maintaining the correct aspect ratio of the graph. Additionally, the algorithm should support real-time updates to provide an accurate and responsive user experience. The transformation of raw data points must be performed quickly, otherwise, there is a risk of data loss. Therefore many parameters are need to be considered when transforming the data.

#### Parameters

- Screen dimensions (height, width)
- Current min and max value in buffer
- Buffer capacity
- $\Delta v$  , where  $\Delta v = \max - \min$  (range of values in current Buffer)

- $\text{coefficient} = \frac{\text{height} - 2 \times \text{shiftH}}{\Delta v}$  (maps raw values to screen height, with margin of shiftH pixels at the top and bottom)
- $\text{step} = \frac{\text{width}}{\text{bufferCap.}}$  (Determines horizontal space for each data point)

All data in the buffer will be updated as shown in the following code snippet:

```

1  for (i in dataTemp.indices) {
2      data[i] = (maxV - dataTemp[i]) * coeff + shiftH
3  }
```

**Code 2:** Mapping ECG-Values to screensized values

The calculation process can be examined in greater detail in the following example, which demonstrates how each data point is transformed and mapped onto the screen.

To keep the example simple and concise, it will be dealt with an array of size 5 and a Coordinate system with a range of (6,6):

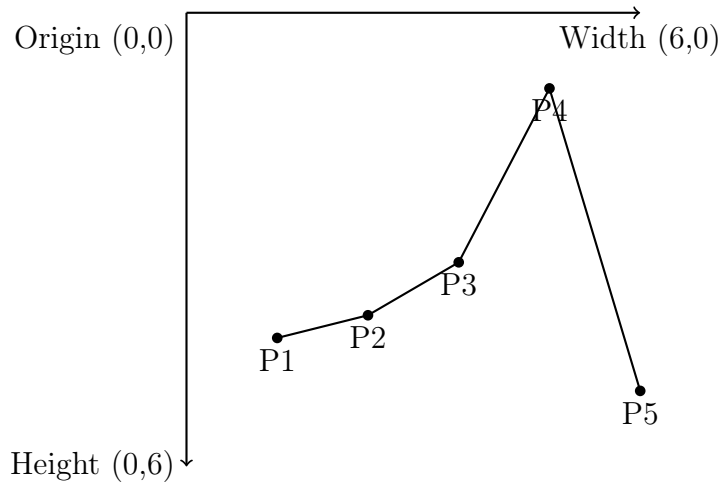
|     |     |     |      |     |
|-----|-----|-----|------|-----|
| 951 | 960 | 980 | 1050 | 930 |
|-----|-----|-----|------|-----|

- Screen dimensions (6,6)
- $\text{minV} = 930$ ,  $\text{maxV} = 1050$
- $\text{BufferCapacity} = 5$
- $\Delta v = 1050 - 930$
- $\text{coefficient} = \frac{6 - 2 \times 1}{120}$  (shiftH = 1 )
- $\text{step} = \frac{6}{5}$

After applying the formula of Code 2 to every element, the newly evaluated Array contains the following data:

4.4
4.0
3.3
1.0
5.0

Resulting in the following drawing:



Resulting in the following Complexity Analysis:

- Data Conversion:  $O(n)$  (for  $n$  data points)
- Scaling & Mapping:  $O(n)$  (for  $n$  data points)
- Overall Complexity:  $O(n)$  linear (for  $n$  data points)

The resulting linear Complexity of  $O(n)$  makes the algorithm efficient for real-time processing. [Liu and Layland \[1973\]](#)

### 3.2. Drawing ECG-Data in realtime

Processing real-time electrocardiographic (ECG) data presents multiple challenges that must be addressed to ensure accurate and efficient visualization. It is helpful to split this rather complex functionality into multiple concerns:

- The sample rate of the microcontroller (1)
- Consume and process incoming data simultaneously (2)
- Datahandling and Drawing in parallel (3)

These issues must be viewed and solved isolated, and subsequently combine each solved problem with the other. It is important to mention that some issues may create restrictions for other issues. Therefore, it is essential to implement one task at a time and connect them all afterward.

### (1) Handle The Sample rate of the microcontroller

The microcontroller sends 10 samples every 8 milliseconds (1000ms = 1s).

$$\frac{1000}{8} = \frac{125}{1} \hat{=} 125Hz$$

Hence the app has to process a sample rate of:  $f(t) = 125 \times t$ , t in seconds. If handling this datastream is not possible, data will be lost.

### (2) Consume and process incoming data at the same time

To consume and process data at the same time, splitting up processing and consuming into two indepent tasks by assigning each task to its own thread, is necessary. Making it possible to operate in paralell.

### (3) Datahandling and drawing in parallel

Thus the drawing entity has to countiously read data, and the datahandler entity receives a continous data stream. The suitable data structure must be able to constantly record new data and retain a certain amount of old data. So that a screen-width drawing can be displayed continuously. In addition, it is important that the data structure is not getting to large, which would lead to visible latencies.

This led to the Ringbuffer beeing the datastructure of choice. Ringbuffers are often used when having to process datastreams. The ringbuffer has a fixed size, thus old data gets overwritten. Therefore, there is no need to worry about the data structure consuming too much memory.

## **3.3. 24-Hour recording**

The functional requirements of the 24-Hour recording are shown in the list below. All of the listed requirements will be explainend in detail in chapter 4.

- Recording mode



- Filechooser concept, to choose multiple files
- Start a recording, and assigning a name
- Abort a recording
- Display 24-Hour long recorded ECG-Data
- Show the current time intervalls
- Zooming into the displayed data, zooming out of the displayed data
- Plot BPM of the current displayed time intervall

### 3.4. Connection via Bluetooth Low Energy

The application is required to establish and maintain a reliable Bluetooth Low Energy (BLE) connection with the hardware. A stable connection is essential for continous data-transfer aswell as for the 24-Hour recording Mode. The BLE entity must be capable of the following items:

- Continous datastream
- Communicate with the application
- Share current the current mode (realtime, or recording)
- Scan for near devices
- split & reassamble to large datasets

## 4. REALIZATION

The first part of this chapter provides an overview of the information flow between each entity, and the tasks of the given entitys. Aswell as an close look into the implementation of the functional requirements, by analysing code snippets, aswell as explaining the usage of the final product. While the second part will explain every necessity to assemble all parts together for a working product.

---

## 4.1. Selected Technologies

"The nature of the application and the required functionality of a software system should influence the choice of methods and tools used during development". [Clarke \[1995\]](#) This ensures that the development process is both efficient and tailored to meet the unique demands of the system. Moreover, aligning tools and methods with the application's requirements can help streamline workflows and reduce potential complexities during implementation. Consequently choosing the right programming language, aswell as IDE is essential to enable efficient development.

### 4.1.1. Programming Language and Framework

When searching for a suitable programming language to program Android apps, names like Java/Kotlin and Dart often appear. Both languages offer great frameworks, such as Flutter for Dart or Jetpack Compose for Kotlin. One aswell as the other are highly effective because they eliminate the limitations of XML-based UI development, enabling a more streamlined, declarative approach that combines UI and logic in a single language. Thus reduces boilerplate code, and enhances performance aswell as developer productivity. The main difference between both frameworks lies in resource management. For example Flutter is superior in cpu usage, but Jetpack Compose consumes less memory. [Kusuma et al. \[2023\]](#) "But since both have their respective advantages and disadvantages, it is impossible to say that one is superior to the other. The choice between Jetpack Compose and Flutter depends on the needs and preferences of the developer. If developers are more familiar with the Kotlin language and want to use new technologies, then Jetpack Compose might be a better choice." [Kusuma et al. \[2023\]](#) With prior practical experience in Java/Kotlin, Jetpack Compose was chosen as the framework for this work.

### 4.1.2. Development Environment

Integrated development Environments play a fundamental role in today's application development, providing developer with tools mandatory to debug, refactor and write code. Making programming more efficient and delightful for programmers. When developing Android Apps with state of the art frameworks like Jetpack Compose, Android Studio stands out as the most suitable option as both Jetpack compose and Android Studio are created by Google. Android Studio is the official IDE for Android Development. With IntelliJ as base it inherits powerful code editor and developer tools, combined with Android specific features. Such as Live Edit (updates running app when code is changed) aswell

as Interactive Preview which allows to test functionality between so called Compose components (Compose will be explained in the later course of this work)<sup>3</sup>, et cetera.

The version of Android Studio used for this thesis is **Ladybug | 2024.2.1**. Which is the latest version as this was written (13.12.2024).<sup>4</sup> Downloading the right version can be crucial for problem-free debugging. As some versions lead to bugs, causing the screen to freeze.<sup>5</sup>

## 4.2. Definiton of Entitys

As shown in table 2 each entity has its scope of dutys, making the final code more modular and consequently maintainable.

**Table 2:** Entiys area of responsibility

| Layer             | Functionality                                                                                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BLEViewModel      | <ul style="list-style-type: none"> <li>- Communication with the microcontroller</li> <li>- Receives datastream</li> <li>- Sends instructions to microcontroller</li> </ul> |
| GraphEcgWidget    | Draws the ECG data in realtime, and displays Beats per minute                                                                                                              |
| Storewrapper      | Converts data from microcontroller into drawble data                                                                                                                       |
| BPM               | Calculates the current Beats Per Minute                                                                                                                                    |
| VitalMonitorWidet | Plots up to 24 Hour long recoreded ECG data and the BPM of the plotted Intervall. Provides zooming functionality.                                                          |
| StorageHandler    | Allows usage of the OS filechooser, to select files.                                                                                                                       |
| MainView          | Collection of visble elements (Widgets), to provide reusability                                                                                                            |
| MainActivity      | Visible to user, contains the complete User Interface. Takes and processes user inputs.                                                                                    |

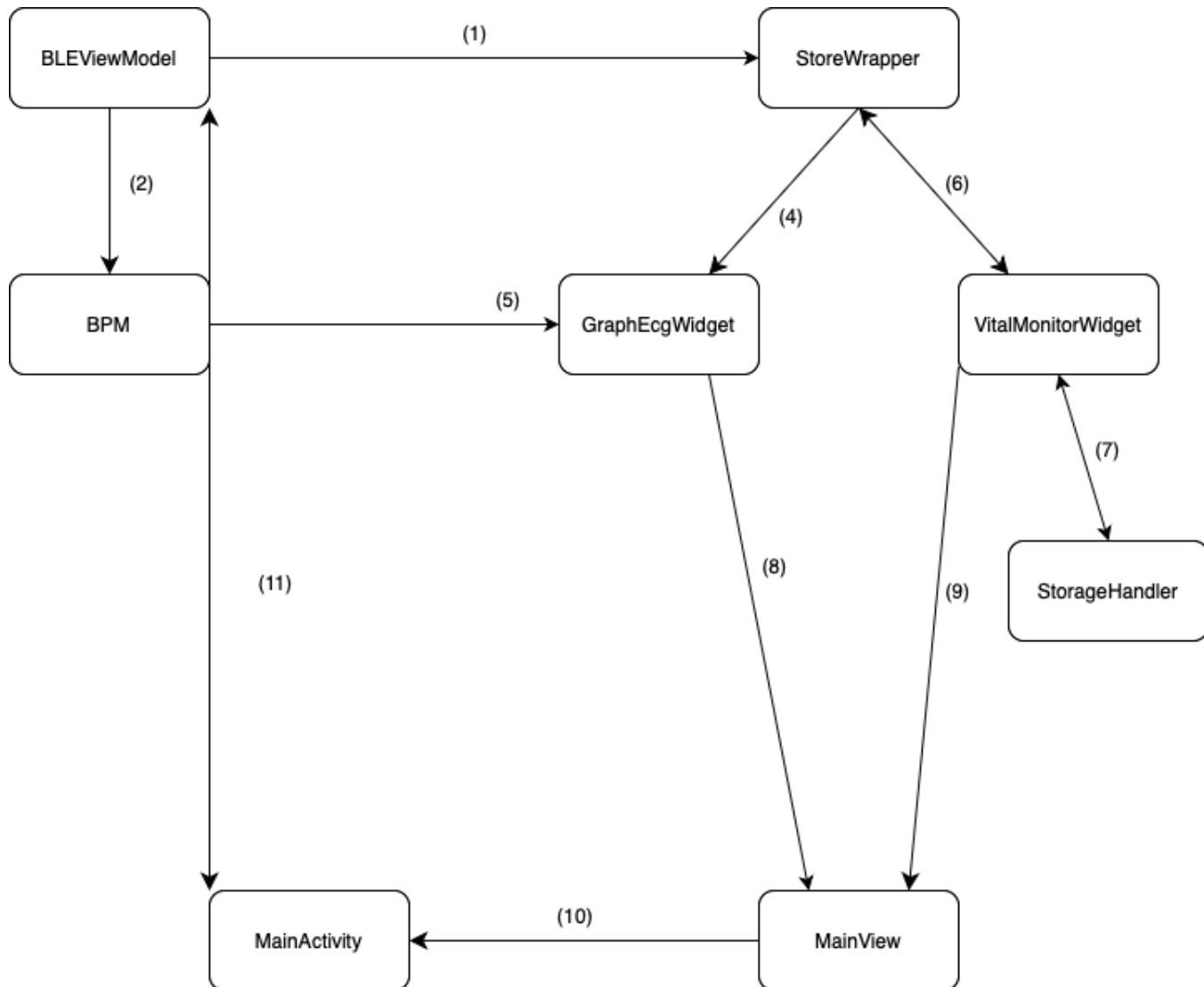
<sup>3</sup><https://developer.android.com/studio/intro>

<sup>4</sup><https://developer.android.com/studio/releases>

<sup>5</sup><https://stackoverflow.com/questions/78125635/processing-classes-for-emulated-method-breakpoints-this-popup-comes-and-loadin>

### 4.3. Inter-Entity Communication Overview

The following diagram shows the Information flow between all entities. The unidirectional arrow represents, unidirectional information flow. On the contrary bidirectional information flow between two entities is represented by an bidirectional arrow.



**Figure 5:** Information-Flow-Diagramm

#### Information Flow and Responsibilities of BLEViewModel

- **BLEViewModel as an Interface**

- Acts as an interface between the controller and the app (4.2).

- 
- Any entity that evaluates recorded data depends on this instance.
  - **Entities Dependent on BLEViewModel**
    - **StoreWrapper**
      - \* Makes the data drawable (1).
      - \* Forwards the data to the **GraphECGWidget** (4).
    - **BPM Instance**
      - \* Calculates the beat frequency (2).
      - \* The result is displayed by the **GraphECGWidget** (5).
  - **Comparison of Information Flows (4) and (6)**
    - Both provide drawable data to widgets.
    - **Difference:**
      - \* **(4) One-way flow** → StoreWrapper receives data directly from the ECG.
      - \* **(6) Bidirectional flow** → Required for reading recorded data from a local file (7) using the **StorageHandler** before sending it to the StoreWrapper.
    - Once mapped by the StoreWrapper, the data is sent back to the **VitalMonitorWidget** (6).
  - **MainView Responsibilities**
    - Integrates **GraphECGWidget** (8) and **VitalMonitorWidget** (9).
    - Manages the visibility of these components.
  - **Extended BLEViewModel Responsibilities**
    - Beyond data exchange, BLEViewModel also:
      - \* Scans and connects to other nearby devices.
      - \* Integrates and manages 24-hour ECG recordings.
    - These functionalities are used by **MainView** and **MainActivity**.
  - **MainView vs. MainActivity**
    - **MainView**
      - \* Constructs components and functionalities for **MainActivity**.
      - \* Provides UI elements, such as buttons for starting a 24-hour ECG recording (10).

### – MainActivity

- \* Acts as a bridge between the **microcontroller** and the **user**.
- \* Handles device scanning, recording initiation, connection establishment, and ECG data display (11).

## 4.4. BLEViewModel

The following subsections explore the logical implementation and functionality of individual entities, using code snippets. Each subsection is divided into a section covering the implementation and another visualizing the code. Framework specific code fragments will be explained in the course of relevant section.

### Dependency versions and Library

Before delving into the implementation, it is essential to note that a library provided by Nordic Semiconductor was employed in the development process.<sup>6</sup> The utilized dependency version are shown in the figure below:

```
implementation("no.nordicsemi.android.kotlin.ble:scanner:1.1.0")
implementation("no.nordicsemi.android.kotlin.ble:client:1.1.0")
implementation("no.nordicsemi.android.kotlin.ble:advertiser:1.1.0")
implementation("no.nordicsemi.android.kotlin.ble:server:1.1.0")
implementation("no.nordicsemi.android:ble:2.4.0")
implementation("org.jetbrains.kotlinx:kotlinx-coroutines-core:1.6.0")
implementation("io.coil-kt:coil-compose:2.2.2")
implementation("androidx.collection:collection:1.4.3")
```

Figure 6: Library Versions

### Scanning devices

The following code snippet demonstrates the implementation of a BLE scanning method used to detect nearby devices. The *startScan* function initializes the scanning process by setting the *scanState* to *SCANNING* and resetting the list of discovered BLE devices. The scanning operation is performed using *BleScanner*, with results processed by *BleScanResultAggregator*. The detected devices are filtered to include only those with the name *"THI-EKG"* and stored in a *MutableState*. The scan runs asynchronously within the *viewModelScope*, and after a delay of 2000 milliseconds, it is terminated by canceling *scanJob* and updating the *scanState* to *DONE*. The snippet also contains two annotations above the function definition, which don't add any functionality, but provide robustness

<sup>6</sup><https://github.com/NordicSemiconductor/Android-BLE-Library>

and readability to the code. Nevertheless, a brief explanation follows to avoid confusion and ensure clarification:

#### @RequiresApi(Build.VERSION\_CODES.UPSIDE\_DOWN\_CAKE)

This annotation ensures that the *startScan* function is only executed on devices running Android API level **34 (Android 14, Upside Down Cake)** or higher. If the function is called on a lower API level, the compiler will issue a warning, and runtime execution may fail.

#### @SuppressWarnings("MissingPermission")

This suppresses lint warnings related to missing Bluetooth permissions (such as *BLUETOOTH\_SCAN* or *ACCESS\_FINE\_LOCATION*). Normally, scanning for BLE devices requires specific permissions, and Android Studio would generate a warning if they are not explicitly checked. However, this annotation tells the lint tool to ignore such warnings, assuming the necessary permissions are already handled elsewhere in the code. Thus the permissions were always granted in the framework of this thesis, these warning can be ignored, and will appear more often.

1 | `private var scanJob: Job? = null`

```

2
3  @RequiresApi(Build.VERSION_CODES.UPSIDE_DOWN_CAKE)
4  @SuppressWarnings("MissingPermission")
5  private fun startScan(context: Context) {
6      scanState.value = ScanState.SCANNING
7      foundBLEDevices.value = mutableSetOf()
8
9      val aggregator = BleScanResultAggregator()
10     scanJob = BleScanner(context).scan()
11     .map { aggregator.aggregateDevices(it) }
12     .onEach {
13         foundBLEDevices.value = it.stream()
14         .filter { it.name == "THI-EKG" }
15         .collect(Collectors.toSet())
16     }
17     .launchIn(viewModelScope)
18
19     viewModelScope.launch {
20         delay(2000)
21         scanJob?.cancel()
22         scanState.value = ScanState.DONE
23     }
24 }

```

**Code 3:** Code for scanning functionality

## Refreshing

Before initiating the BLE scan, the function checks for two critical permissions:

- **BLUETOOTH\_SCAN:** This permission is required to scan for nearby BLE devices.
- **BLUETOOTH\_CONNECT:** This permission is necessary to establish a connection with discovered BLE devices.

The function uses `ActivityCompat.checkSelfPermission` to verify whether these permissions have been granted by the user. This step is crucial for ensuring that the application complies with Android's permission model, which mandates explicit user consent for accessing sensitive device features. Conditional Execution: If both permissions (**BLUETOOTH\_SCAN** and **BLUETOOTH\_CONNECT**) are granted, the function proceeds to call `startScan(context)`.

```

1  @RequiresApi(Build.VERSION_CODES.UPSIDE_DOWN_CAKE)
2  fun refreshBLE(context: Context) {
3      if (ActivityCompat.checkSelfPermission(
4          context,

```



```

5      Manifest.permission.BLUETOOTH_SCAN
6      ) == PackageManager.PERMISSION_GRANTED &&
7      ActivityCompat.checkSelfPermission(
8      context,
9      Manifest.permission.BLUETOOTH_CONNECT
10     ) == PackageManager.PERMISSION_GRANTED
11     ) {
12         startScan(context)
13     }
14 }

```

**Code 4:** Code for refresh functionality

## Connect

The provided BLE implementation is responsible for establishing a connection with a BLE-enabled device, receiving real-time data from specific BLE characteristics, and processing the incoming data to extract and analyze electrocardiographic (ECG) information. The functionality can be divided into the following key components:

### **1. BLE Connection Setup (Line 4-10)**

The function `connect` initializes the connection with a given BLE device. Using `ClientBLEGatt.connect`, a GATT connection is established, and the discovered services and characteristics are retrieved. The relevant characteristics include:

*signalCharacteristic*: Used for receiving real-time electrocardiographic data.

*appCommunicationCharacteristic*: Used for bidirectional communication with the BLE device.

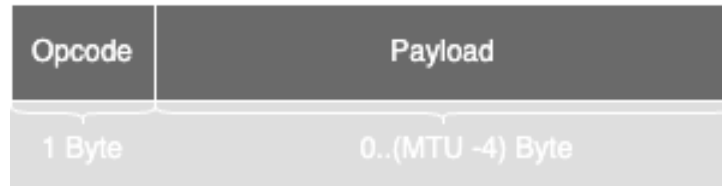
### **2. Receiving and Processing Data from the Signal Characteristic**

The `signalCharacteristic`, continuously provides data in the form of a `ByteArray`. Each received data packet undergoes the following procedure:

The received byte array is first converted into a list of integers. To enable concurrent processing, the data is then transmitted through a channel (`channel.send(values)` in line 15). A coroutine, executed by `receiveData(viewModelScope)`, retrieves the streamed data from the channel and passes it to `StoreWrapper.processDataList(datastream)`. (Since coroutines and channels play a central role in this and other implementations, this chapter is followed by a brief introduction to these Kotlin-specific topics.) The processed data is also added to the `BPM.currList` buffer for frequency analysis. Once the buffer exceeds 7500 samples, the QRS peak detection algorithm (Pan-Tompkins) is applied to identify heartbeats.

### **3. Handling Application-Level BLE Communication**

During their bidirectional communication, both the microcontroller and the wirelessly connected device operate as senders and receivers. To ensure the correct operation, it is essential that the received data is accurately interpreted. This can be achieved by defining an Opcode for every desired behavior, which is contained in every sent packet. This allows incoming data packets to be classified based on their Opcode. As a result, each received packet can be assigned a meaning and subsequently processed accordingly.



**Figure 7:** Packetstructure with Opcode (Source: Stefan Lindner)

Furthermore the application can trigger actions in the microcontroller by sending a response packet. For example, it can start the recording mode.



**Figure 8:** Response Packet (Source: Stefan Lindner)

Therefore each received notification undergoes:

Validation using `ResponseChecker.check`. And packet reconstruction via `PacketReassembler.reassemble`, because some data is send fragmented by the microcontroller since the payload would be to large. Consequently the app needs to reassemble these packages to interpret them correctly.

The check function aswell as the reassemble function will be explained in the following.

#### 4. Validating Packages (ResponseChecker)

The `check()` function verifies whether a given packet corresponds to an expected response from the microcontroller. By verifying the expected Opcode. The verification consists of multiple steps:

##### 1. Initial Length Check

If the byte array is too short ( $\leq 2$  bytes), it cannot be a valid response and is immediately discarded.

## 2. Packet Type Validation

Calls ‘isResponsePacket(byteArray)’, a function that determines whether the received data is a response packet. If the data is not a response packet, the function exits early.

## 3. Opcode Verification

Calls ‘checkForExpectedOPCode(byteArray)’, which checks whether the response contains the expected operation code. If the opcode does not match the expected one, the function exits early.

## 4. Cancelling the Timeout Mechanism

If the response is valid, the current timeout timer (‘currentTimer’) is canceled, ensuring that the system does not time out while waiting for the response.

## 5. Extracting and Processing the Response Message

The actual response message is extracted from the byte array starting at index 2. The message is decoded as a UTF-8 string.

## 6. Notifying the Listener

If a ‘responseReceivedListener’ is set, it is triggered with the received byte array. The listener is likely responsible for further processing or handling the response.

## 7. Resetting the Expected Response

The variable ‘toCheck’, which presumably stores the expected response for comparison, is reset to ‘null’, indicating that the response-checking process has completed.

This function ensures that only valid and expected responses are processed in a BLE communication system. By implementing multiple validation checks (length, packet type, opcode), it prevents erroneous or irrelevant data from triggering unintended behavior.

```
1
2      fun check(byteArray: ByteArray) {
3
4          if(byteArray.size <= 2) {
5
```

```

6         return
7     }
8
9     if(isResponsePacket(byteArray).not()){
10
11         return
12     }
13
14     expected opcode
15     if(checkForExpectedOPCode(byteArray).not()){
16
17         return
18     }
19
20     this.currentTimer!!.cancel()
21
22
23     val message = String(byteArray.sliceArray(2 until byteArray.
24         size), Charsets.UTF_8)
25
26     if(this.responseReceivedListener != null) {
27         this.responseReceivedListener!!(this.toCheck!!, byteArray)
28     }
29
30
31     this.toCheck = null
32 }

```

**Code 5:** Code to validate packages

## 5. Reassembling Packages (ResponseChecker)

This function is responsible for reconstructing fragmented data packets. It ensures that data belonging to the same operation code (opcode) is sequentially assembled and processed once the transmission is complete.

### Initial Validation:

If the received byte array (data) is too short (less than 2 bytes), it is ignored as it is not a valid packet.

### Opcode Relevance Check:

The function checks if the opcode of the received packet is part of a predefined set of relevant opcodes (Constants.RELEVANT\_OPCODES). If the opcode is not relevant, the packet is ignored.

### Handling Unexpected Opcodes:

If a packet with a different opcode arrives while another one is still being processed (`currentOpCode` is not null), an exception is thrown. This prevents interleaved or mixed opcode sequences from corrupting the reassembly process.

Starting a New Message:

If no opcode is currently being processed (`currentOpCode == null`), the function assumes this packet marks the start of a new message and assigns its opcode to `currentOpCode`.

Appending Data:

The function extracts the payload (excluding the first two bytes, which likely contain metadata such as the opcode) and appends it to the `dataStr` buffer.

Completion Check & Listener Notification:

If the packet indicates it is the last fragment (`data.isLast()`) and a listener (`finishedListener`) is set, the listener is notified with the assembled message. After notifying the listener, the function resets the `dataStr` buffer and `currentOpCode` to prepare for the next message.

```
1
2
3  fun reassemble(data: ByteArray) {
4
5      if(data.size < 2) {
```

```

6
7         return
8     }
9
10    if(Constants.RELEVANT_OPCODES.contains(data.opCode()).not()){
11
12        return
13    }
14
15    if(currentOpCode != null && this.currentOpCode != data.opCode()) {
16
17        throw RuntimeException("Unexpected OpCode received!")
18    }
19
20    if(currentOpCode == null) {
21
22        currentOpCode = data.opCode()
23    }
24
25
26    dataStr += String(data.sliceArray(2 until data.size), Charsets.
        UTF_8)
27
28    if(data.isLast() && finishedListener != null){
29
30        finishedListener!!(this.currentOpCode!!, this.dataStr)
31
32
33        this.dataStr = ""
34        this.currentOpCode = null
35    }
36 }

```

Code 6: Packet reassembler code

## 6. Additional Commands

The implementation also includes:

Sending an MTU request (requestMtu(512)) to maximize packet size for efficient data transmission. Checking if the application is in live mode or in 24-Hour mode by calling fetchCurrentMode(). Also fetching all recorded ECG's from the microcontroller ListEKGS().

---

### Channels and Coroutines

The *channel* serves as a concurrent data pipeline between the BLE data stream and the processing logic. It ensures that the incoming data from the BLE device is processed in a structured, asynchronous manner without blocking the main execution thread. By leveraging coroutines, the channel allows efficient real-time data handling while preventing performance bottlenecks. [JetBrains \[2024a\]](#)

A *coroutine* in Kotlin is a lightweight, non-blocking way to handle asynchronous programming. It allows to write code that can be paused and resumed without blocking the main thread, making it ideal for handling tasks like network requests, BLE communication, or large computations without freezing the UI. [JetBrains \[2024b\]](#)

```
1  @SuppressLint("MissingPermission")
2  fun connect(context: Context, device: ServerDevice) {
3      this.viewModelScope.launch {
4          val connection = ClientBleGatt.connect(context, device,
5              this@BLEViewModel.viewModelScope )
6          this@BLEViewModel.connection.value = connection
7          val services = connection.discoverServices()
```

```

7      val bleService = services.findService(UUID.fromString("e28e00e9
      -b79a-479a-a6f1-21e8f73236e1"))
8      val signalCharacteristic = bleService!!.findCharacteristic(UUID
      .fromString("23b85b26-e7cb-421c-9731-1b4ca2d2a849"))
9      val appCommuncationCharacteristic = bleService!!.
      findCharacteristic(UUID.fromString("563a2846-1dc7-11ef
      -9262-0242ac120002"))
10     this@BLEViewModel.appCommuncationCharacteristic =
      appCommuncationCharacteristic!!

11
12     signalCharacteristic!!.getNotifications().onEach {
13         val values = byteArrayToIntArray(it.value).toList()
14         startTime = System.currentTimeMillis()
15         channel.send(values)
16         receiveData(viewModelScope){dataList ->
17             StoreWrapper.processDataList(dataList)
18             BPM.currList.value += dataList
19             if(BPM.currList.value.size >= 7500){
20                 var qrsPeaks = BPM.panTompkins(BPM.currList.value).
                    size
21
22                 BPM.currList.value.forEach {
23
24                 }
25
26                 BPM.currList.value = BPM.currList.value.subList
                    (750, 7500-1)
27                 BPM.countPeak = qrsPeaks
28                 qrsPeaks = 0
29             }
30         }
31         paketCounter++
32     }.launchIn(viewModelScope)
33     appCommuncationCharacteristic!!.getNotifications().onEach {
34         ResponseChecker.check(it.value)
35         PacketReassembler.reassemble(it.value)
36
37     }.launchIn(viewModelScope)
38     connection.requestMtu(512)
39     fetchCurrentMode()
40     listEKGS()
41 }
42 }

```

Code 7: Connect functionality code



## 4.5. Storewrapper

The StoreWrapper class serves as a central component for processing and managing electrocardiographic (ECG) data in real-time. It is responsible for converting raw data received from the BLE signal characteristic into a format suitable for visualization. This transformation ensures that the incoming stream of ECG values can be efficiently rendered within the application's graphical interface. A key feature of the StoreWrapper class is its mapping functionality, which translates raw numerical values into drawable data points. This conversion is essential for accurately representing the ECG signal on-screen. Additionally, the class incorporates a ringbuffer, which provides a mechanism for storing and handling continuous data streams. The ringbuffer ensures that only the most recent samples are retained, enabling a sliding window approach for real-time signal visualization and processing.

### Preparing ECG-Data

The provided code snippet is responsible for transforming raw electrocardiographic (EKG) data into a format suitable for visualization across different screen sizes. This transformation ensures that the signal is correctly scaled and displayed within the graphical interface, regardless of resolution or aspect ratio. The function achieves this with a time complexity of  $O(n)$ , making it efficient for real-time applications.

The function begins by initializing the shiftH parameter and preparing a mutable list (data) for storing processed values. If an external data source is provided, it is converted from a generic format into a list of Double values. Otherwise, an empty list is created.

### Screen Size and Scaling Factors

The width and height of the target display area are extracted from the size parameter. Additionally, the function determines the minimum (minV) and maximum (maxV) values within the buffer to establish the dynamic range of the data. If all values are equal, an artificial spread is introduced to prevent division by zero.

### Normalization and Scaling

The difference between maxV and minV (dv) is calculated to determine the range of data values. A step size (step) is then computed based on the available screen width and the buffer capacity, ensuring that data points are evenly distributed across the display. The

coeff value is derived to scale the signal to fit within the available height while accounting for a vertical shift (shiftH).

### Data Mapping for Graph Rendering

A temporary list (dataTemp) holds the normalized data, which is then mapped to a new list (data). The mapping follows the formula:

$$data[i] = (maxV - dataTemp[i]) \times coeff + shiftH$$

This equation ensures that the data is scaled appropriately for graphical representation. The function iterates over the dataset in a single pass (O(n) complexity), making it highly efficient for real-time applications. This guarantees smooth and responsive rendering, even with large data sets.

```
1 fun prepareData(size: Size, shiftH: Double, dataFromFile:
2   MutableList<Any>? = null): List<Double> {
3   this.SHIFTH = shiftH
4   var data: MutableList<Double>? = null
5   if( dataFromFile != null ){
      data = dataFromFile.mapNotNull {
```

```

6         (it as? String)?.toDouble()
7     }.toMutableList()
8 }else{
9     data = ArrayList()
10 }
11 val width = size.width.toDouble()
12 val height = size.height.toDouble()
13 var minV = min //lowest value in buffer
14 var maxV = max //highest value in buffer
15 if (minV == maxV) {
16     minV = minV / 2
17     maxV = maxV + minV / 2
18 }
19 val dv = maxV - minV
20 step = width / buffer_.capacity()
21 val coeff = (height - 2 * shiftH) / dv
22
23 val dataTemp =
24 if (_mode == GraphMode.overlay) utils_.dataSeriesOverlay(
25     buffer_) else utils_.dataSeriesNormal(
26     this
27 )
28 data = ArrayList(Collections.nCopies(dataTemp.size, 0.0))
29 //return data
30 if (data.isEmpty() || dataTemp.isEmpty()) {
31     return data
32 }
33 for (i in dataTemp.indices) {
34     data[i] = (maxV - dataTemp[i]!!) * coeff + shiftH
35 }
36 return data
37 }

```

Code 8: Code for mapping data on the screen

## Preparing the Drawpath

To achieve a smooth and natural representation of the ECG waveform, it is essential to create two separate paths: one for the data points before the current write position in the buffer and one for the data points after. This ensures a seamless transition between old

and new data, preventing visual gaps and making the waveform appear continuous, just like a real ECG signal.

The function `preparePathBefore(data: List<Double>)` constructs the path for the portion of the data preceding the current write index of the buffer. It determines the correct starting index and iterates through the data points up to the write position, generating a Path object that maps the signal to drawable coordinates.

Conversely, the function `preparePathAfter(data: List<Double>)` handles the data following the write index. It ensures that the path continues from the last drawn point, maintaining a smooth flow of the signal. By splitting the waveform into these two sections, the transition between old and new data is visually imperceptible, preserving the real-time appearance of the ECG signal.

This approach is crucial for maintaining a continuous and realistic ECG display, as it eliminates abrupt jumps or gaps in the graph, ensuring that the visualization remains fluid and accurately represents the ongoing signal dynamics.

```
1 fun preparePathBefore(data: List<Double>): Path {  
2     var idx_ = buffer_.writeIndex() - 1  
3     if (idx_ >= data.size) {  
4         idx_ = data.size - 1  
5     }  
6     }  
7     val idx = if (idx_ < 0) 0 else idx_  
8     val path = Path()
```

```

9      path.moveTo(0f, data[0].toFloat())
10     for (i in 1 until idx - 1) {
11         path.lineTo((i * step).toFloat(), data[i].toFloat())
12     }
13     return path
14 }
15
16 fun preparePathAfter(data: List<Double>): Path {
17     var idx_ = buffer_.writeIndex()
18     if (idx_ >= data.size) {
19         idx_ = data.size - 1
20     }
21     val idx = if (idx_ < buffer_.capacity() - 1) idx_ else buffer_.
        capacity() - 1
22     val path = Path()
23     path.moveTo((idx * step).toFloat(), data[idx].toFloat())
24     for (i in idx + 1 until data.size) {
25         path.lineTo((i * step).toFloat(), data[i].toFloat())
26     }
27     return path
28 }

```

Code 9: Code to prepare the ECG-Path

## 4.6. GraphEcgWidget

This code implements a real-time ECG visualization. The code renders ECG data as a graph, calculates the heart rate (BPM), and classifies it into different heart rate categories. Since the code of this class is very extensive, some code snippets are presented in abbreviated form.

### Displaying Heart Rate

The function *ECGValues(frequency: Int)* displays the current BPM and classifies it into one of three categories:

- **Bradycardia** (heart rate < 60 BPM)
- **Normal Frequency** (heart rate 60-99 BPM)
- **Tachycardia** (heart rate  $\geq$  100 BPM)

```

1 @Composable
2 fun ECGValues(frequency: Int){
3     ...
4     when {
5         frequency > 0 && frequency < 60 -> { /* Bradycardia */ }
6         frequency in 60..99 -> { /* Normal frequency */ }
7         frequency >= 100 -> { /* Tachycardia */ }
8     }
9     ...
10 }

```

**Code 10:** Code to classify current heartrate

### Continuous ECG-Graph

The overlay graph ensures that the ECG waveform remains continuous by storing and displaying previous data points alongside the latest ones, rather than just showing the most recent signal. This prevents data loss and allows a smooth, uninterrupted visualization of the heartbeat over time. The `drawOverlayGraph()` function is responsible for rendering a continuous ECG graph by maintaining both past and current data. Instead of only displaying the most recent values, it preserves previously recorded data, ensuring that the graph remains complete and does not lose old information.

First, the function retrieves two paths:

`pathBefore`, which contains the past ECG data, ensuring historical values remain visible. `pathAfter`, which holds the newest ECG data points, representing the latest heartbeat progression. To ensure a clear visualization, the function checks whether there is a stored point from the latest ECG reading.

Then, both paths are drawn sequentially:

`pathBefore` is drawn in red with a thin stroke, representing the past ECG signal. `pathAfter` is drawn with a slightly thicker red stroke, making the latest ECG values more prominent. Finally, the function highlights the most recent ECG data point by drawing a red dot at the latest position, helping users identify the real-time measurement. By continuously displaying both past and new data, the overlay graph creates a seamless ECG visualization, allowing for smooth tracking of the heart's electrical activity over time.

```

1 fun drawOverlayGraph() {
2     val pathBefore = if (storeWrapper.pathBefore == null) Path()
3                       else storeWrapper.pathBefore!!
4     val pathAfter = if (storeWrapper.pathAfter == null) Path() else
5                     storeWrapper.pathAfter!!

```

```

5         if (storeWrapper.point != null) {
6             drawRect(
7                 color = Color.Transparent,
8                 topLeft = Offset(storeWrapper.point!!.x, 0f),
9                 size = Size(width, height),
10                style = Fill
11            )
12        }
13
14        drawPath(
15            path = pathBefore,
16            color = Color.Red,
17            style = Stroke(width = 1f),
18        )
19
20        drawPath(
21            path = pathAfter,
22            color = Color.Red,
23            style = Stroke(width = 2f),
24        )
25
26        if (storeWrapper.point == null) return
27
28        drawCircle(
29            center = Offset(storeWrapper.point!!.x, storeWrapper.point!!.y)
30            ,
31            radius = 6f,
32            color = Color.Red
33        )
34    }

```

**Code 11:** Code to draw the prepared Overlaygraph

### Real-time graph update

The *doneClick* function is responsible for starting or stopping the ECG data visualization when the user interacts with the graph. It first toggles the *isCoroutineRunning* flag to determine whether the coroutine should be started or stopped. It then calls *toggleCoroutine*, passing the new state along with the coroutine scope and the number of cycles. Within the coroutine, the function updates *startTime* with the current system time to measure

performance and increments `currentCounter` to update the UI. If the counter is greater than zero, it prepares the drawing by calling `storeWrapper.prepareDrawing`, passing the screen width and height along with a fixed scaling factor. This ensures that the visualization area is correctly set up for rendering. Finally, the function updates `heartFrequency` by retrieving the latest BPM value from `BPM.countPeak`, ensuring that the displayed heart rate is continuously updated with the latest processed data. This function plays a crucial role in managing real-time ECG visualization, controlling when data processing runs and ensuring the graph updates dynamically as new data arrives.

```

1      fun doneClick(cycles: Int) {
2          val width = screenWidth
3          val height = screenHeight
4          isCoroutineRunning = !isCoroutineRunning
5          toggleCoroutine(isCoroutineRunning, coroutineScope, cycles) {
6              counter ->
7                  currentCounter = counter + 1
8                  if(counter > 0){
9                      storeWrapper.prepareDrawing(android.util.Size(width.
10                         toInt(), height.toInt()), 32.toDouble())
11                  }
12              heartFrequency = BPM.countPeak
13          }
14      }

```

**Code 12:** Code to toggle drawing per userinput

The `toggleCoroutine` function controls the execution of a coroutine that processes ECG data. It takes a boolean `isRunning` to determine whether the coroutine should start or stop, a `coroutineScope` to manage coroutine execution, an integer `cycles`, and a callback function that receives an integer value. When `isRunning` is true, a new coroutine is launched, and a channel is created to continuously send incremented counter values at an interval of 8 milliseconds. These values are then passed to the callback function, ensuring that the ECG data is updated in real time. If `isRunning` is false, all active child coroutines within the given scope are canceled, stopping data processing. This function ensures controlled and efficient execution of real-time ECG data handling, starting and stopping dynamically based on user interaction.

```

1      fun toggleCoroutine( isRunning: Boolean, coroutineScope:
2          CoroutineScope, cycles: Int, callback: (Int) -> Unit) {
3          coroutineScope.launch {
4              if (isRunning) {

```



```

4         val channel = Channel<Int>()
5         launch {
6             while (true) {
7                 channel.send(counter++)
8                 delay(8)
9             }
10        }
11        for(value in channel){
12            callback(value)
13        }
14    } else {
15        coroutineScope.coroutineContext.cancelChildren()
16    }
17 }
18 }

```

**Code 13:** Code to start/stop the drawing routine

#### 4.7. VitalMonitorWidget

The VitalMonitorWidget serves as a comprehensive visualization tool for mapping recorded 24-hour ECG data alongside the corresponding BPM values. It incorporates extensive functionality from the GraphECGWidget, to ensure smooth and accurate data representation. Additionally, it features interactive zooming functionality, allowing users to focus on smaller time windows where BPM values align precisely with the corresponding ECG waveforms. The data displayed within the widget is sourced from the file chooser, which is discussed in detail later in the thesis.

##### Draw recorded BPM

The drawFlowingGraph function is responsible for rendering the BPM graph. It first checks if there is available data and ensures the counter is greater than zero before proceeding. It then determines the minimum and maximum values within the data set to establish the graph's vertical range. The function calculates the indices defining the portion of the data to be displayed, ensuring the graph remains within the zoomed-in section while allowing for user interactions such as tapping and zooming. A sublist of the BPM data is extracted, and if all values are equal, adjustments are made to ensure proper scaling.

```
1 fun drawFlowingGraph() {  
2     if(currentCounter > 0 && !dataFromFileBPM.isEmpty() && num == 0){  
3         var minV = dataFromFileBPM.minOf { it.toString().toFloat() }  
4         var maxV = dataFromFileBPM.maxOf { it.toString().toFloat() }  
5         fromIndexBPM = if (tappedPointerIndexBPM - zoomValue <= 0) 0  
6             else tappedPointerIndexBPM-zoomValue  
7         toIndexBPM = if (tappedPointerIndexBPM + zoomValue >=  
8             dataFromFileBPM.size) dataFromFileBPM.size else if (
```

```

        tappedPointerIndexBPM == 0) dataFromFileBPM.size else
        tappedPointerIndexBPM + zoomValue
7   fromTime = fromIndexBPM
8   toTime = toIndexBPM
9   currentBPMList = dataFromFileBPM.subList(fromIndexBPM, toIndexBPM)
10
11
12
13   if (minV == maxV) {
14       minV = minV / 2
15       maxV = maxV + minV / 2
16   }
17
18   val dv = maxV-minV
19   step = width / currentBPMList.size
20   val coeff = (convertDpToFloat(density,150.dp) - 2 * shiftH) / dv
21   var path = Path().apply{
22       currentBPMList.forEachIndexed{i, point ->
23           var preparedPoint= 0.0f
24           if (maxV != null) {
25               preparedPoint = ((maxV - point.toString().toFloat()) *
26                   coeff + shiftH).toFloat()
27           }
28           if(i == 0 ){
29               moveTo((i*step).toFloat(), preparedPoint.toFloat())
30           }else{
31               lineTo((i*step).toFloat(),preparedPoint.toFloat())
32           }
33           currentCounter++
34       }
35   }
36   drawPath(
37   path = path,
38   color = Color.Red,
39   style = Stroke(width = 2f), // Stroke width
40   )
41   ...
42 }

```

Code 14: Code to draw the recorded BPM-Data

### Draw recorded ECG-Data

The second part of the drawFlowingGraph function is responsible for rendering ECG data

with zooming functionality, and detailed visualization of the recorded signals. It first ensures that the ECG data is available and that the current counter and tapped pointer values are valid. The function determines the range of data points to be displayed by defining `fromIndex` and `toIndex`, ensuring that the graph remains within the available dataset while focusing on a smaller segment around the tapped pointer. By dynamically selecting the data range based on user interaction and scaling the values appropriately, this function enables smooth zooming functionality. Users can focus on specific sections of the ECG signal while maintaining accurate and proportional visualization of the waveform.

```

1      ...
2
3      if( num == 1 && !dataFromFilePoints.isEmpty() && currentCounter > 0
4          && tappedPointer > 0){
5          var fromIndex = if (tappedPointer-500 <= 0)    0  else
6                          tappedPointer-500
7          var toIndex = if (tappedPointer+ 500 >= dataFromFilePoints.size
8                          ) dataFromFilePoints.size else tappedPointer+ 500
9          var minV = dataFromFilePoints.minOf { it.toString().toFloat() }
10         var maxV = dataFromFilePoints.maxOf { it.toString().toFloat() }
11         if (minV == maxV) {
12             minV = minV / 2
13             maxV = maxV + minV / 2
14         }
15         val dv = maxV-minV
16         step = width / dataFromFilePoints.subList(fromIndex, toIndex).
17               size
18         val coeff = (convertDpToFloat(density,150.dp) - 2 * shiftH) /
19                   dv
20         var path2 = Path().apply{
21             dataFromFilePoints.subList(fromIndex, toIndex).
22                 forEachIndexed { i, point ->
23
24                 var preparedPoint = 0.0f
25
26                 if (maxV != null) {

```

```

21         preparedPoint = ((maxV - point.toString().toFloat()
22             ) * coeff + shiftH).toFloat()
23     }
24     if(i == 0 ){
25         moveTo((i*step).toFloat(), preparedPoint.toFloat())
26     }else{
27         lineTo((i*step).toFloat(),preparedPoint.toFloat())
28     }
29     currentCounter++
30 }
31
32 }
```

**Code 15:** Code to draw the recorded ECG-Data

### Zooming functionality

This code implements a zooming functionality for the ECG visualization based on user tap interactions. The zooming behavior is controlled through single and double taps, which adjust the zoom level and update the reference point for the displayed data. When a double-tap occurs, the zoom value increases by 150, effectively zooming into a more detailed view of the ECG data. The tappedPointerIndexBPM is then set to the center of the BPM data list, ensuring that the zoom focuses on the middle of the recorded data. When a single tap is detected, the function adjusts the zoom level while updating the tapped-Pointer position based on the x-coordinate of the tap event. The relativeTap variable normalizes the tap position within the screen width, allowing for accurate mapping between the screen coordinates and the data indices. The currentTappedPointerIndexBPM is then calculated relative to the currently displayed BPM list, and tappedPointerIndexBPM is updated to reflect the corresponding index in the original dataset. The zoom value is dynamically adjusted, decreasing by 30 with each tap until it reaches a minimum of 15, ensuring a smooth and controlled zooming experience. Additionally, the currentCounter is incremented, triggering updates in the visualization. This approach allows users to interactively zoom in and out of the ECG graph by tapping, providing a more detailed or broader view of the data as needed.

```

1      .pointerInput(Unit) {
2          detectTapGestures (onDoubleTap = {
3              zoomValue += 150
4              tappedPointerIndexBPM = dataFromFileBPM.size/2
5
6
7          }, onTap = {point ->
8              if (zoomValue >= 150){
9                  zoomValue = 150
10             }
11             tappedPointer = point.x.toInt()
12             var relativeTap = tappedPointer.toFloat() / 720
13             currentTappedPointerIndexBPM = (relativeTap *
14                 currentBPMList.size).toInt()
15             tappedPointerIndexBPM = currentTappedPointerIndexBPM +
16                 fromIndexBPM
17             lastTappedPointerIndexBPM = tappedPointerIndexBPM
18             zoomValue = if (zoomValue <= 30) 15 else zoomValue - 30
19             currentCounter++
20         })

```

**Code 16:** Code for zooming in and zooming out via tap/double tap

#### 4.8. Calculating BPM (Pan-Tompkins Algorithm)

This code implements the Pan-Tompkins algorithm for detecting QRS complexes in an ECG signal. The algorithm processes the raw ECG data through multiple stages of filtering, differentiation, squaring, integration, and peak detection to identify heartbeats accurately. The `panTompkins` function in code 17 serves as the main processing pipeline.

It first applies a bandpass filter, which is a combination of a low-pass and a high-pass filter, to remove noise and unwanted frequency components from the raw ECG data. The filtered signal then undergoes differentiation, enhancing the slope of the QRS complex. Next, the squared transformation is applied to amplify higher amplitude variations, making QRS peaks more distinguishable. The resulting signal is further processed using moving window integration, which smooths the waveform and highlights the QRS complexes by averaging values over a short period. Finally, peak detection is performed. A dynamic threshold is determined based on the signal's overall amplitude, and only peaks that exceed this threshold and maintain a minimum interval (to prevent false detections) are identified as QRS complexes. The `lowPassFilter` and `highPassFilter` functions implement basic digital filters to isolate the frequency range of interest, while `bandpassFilter` applies them sequentially. The `differentiate` function enhances rapid signal changes characteristic of QRS complexes. The `movingWindowIntegration` function smooths the processed signal by averaging values over a predefined window. The `detectPeaks` function then scans the processed data for local maxima that exceed a threshold and are sufficiently spaced apart (at least 200 milliseconds), ensuring only valid QRS peaks are detected. The algorithm ultimately returns a list of QRS peak positions, which can be used for heart rate analysis and further ECG interpretation. [Pingale \[2014\]](#)

```

1      fun panTompkins(ecgData: List<Int>): List<Int> {
2
3          val filteredData = bandpassFilter(ecgData)
4          val derivativeData = differentiate(filteredData)
5          val squaredData = derivativeData.map { it.toDouble().pow(2).
              toInt() }
6          val integratedData = movingWindowIntegration(squaredData)
7          return detectPeaks(integratedData, SAMPLE_RATE)
8      }
9      fun lowPassFilter(signal: List<Int>): List<Int> {
10         var y = MutableList(signal.size) { 0 }
11         for (n in signal.indices) {
12             if (n >= 12) {
13                 y[n] = 2 * y[n - 1] - y[n - 2] + signal[n] - 2 * signal
14                     [n - 6] + signal[n - 12]
15             } else {
16                 0
17             }
18         }
19         return y
20     }
21     fun highPassFilter(signal: List<Int>): List<Int> {
22         var y = MutableList(signal.size) { 0 }
23         for (n in signal.indices) {

```

```

24         if (n >= 32) {
25             y[n] = y[n - 1] - (signal[n] / 32) + signal[n - 16] -
                signal[n - 17] + (signal[n - 32] / 32)
26         } else {
27             0
28         }
29     }
30     return y
31 }
32
33 fun bandpassFilter(ecgData: List<Int>): List<Int> {
34     var lp_filtered = lowPassFilter(ecgData)
35     var hp_filtered = highPassFilter(lp_filtered)
36     return hp_filtered
37 }
38
39 fun differentiate(data: List<Int>): List<Int> {
40     var y = MutableList(data.size) { 0 }
41     for (n in data.indices){
42         if (n >= 4) {
43             y[n] = (2 * data[n] + data[n - 1] - data[n - 3] - 2 *
                data[n - 4]) / 8
44         }
45         else{
46             y[n] = 0
47         }
48     }
49     return y
50 }
51
52 fun movingWindowIntegration(data: List<Int>): List<Int> {
53     val result = mutableListOf<Int>()
54     for (i in data.indices) {
55         val start = maxOf(0, i - WINDOW_SIZE / 2)
56         val end = minOf(data.size, i + WINDOW_SIZE / 2)
57         val average = data.subList(start, end).average()
58         result.add(average.toInt())
59     }
60     return result
61 }
62
63
64 fun detectPeaks(signal: List<Int>, fs: Int): List<Int> {
65     val threshold = signal.sum() / signal.size
66     val peaks = mutableListOf<Int>()
67     val rrInterval = (0.2 * fs).toInt()

```



```

68     var lastPeak = -rrInterval
69
70     for (n in 1 until signal.size - 1) {
71         if (signal[n] > threshold && signal[n] > signal[n - 1] &&
72             signal[n] > signal[n + 1]) {
73             if (n - lastPeak >= rrInterval) {
74                 peaks.add(n)
75                 lastPeak = n
76             }
77         }
78     }
79     return peaks
80 }

```

Code 17: Pan-Tompkins algorithm

#### 4.9. StorageHandler

This function provides a file selection mechanism using Jetpack Compose and Android's Storage Access Framework, allowing users to pick multiple files from their device. It utilizes `rememberLauncherForActivityResult` to open a file picker and handle the selection result. When a user clicks the "Select doc" button, the launcher is triggered, allowing them to choose files. The selected file URIs are processed, and based on their type, they are categorized into a mutable map: Binary files (.bin) are read using `readShort(file_)` and stored under the "bin" key. Metadata files (if the file contains "start") are split by ";" and stored under "meta". Other text files (e.g., BPM data) are also split by ";" and stored under "bpm". This function returns a map containing the parsed data, which can later be used for further processing, such as visualizing ECG and BPM data.

```

1 fun PickDocument(): MutableMap<String, Any>{
2
3     ...
4

```

```

5
6 contract= ActivityResultContracts.OpenMultipleDocuments(){
7
8     result.value = it
9 }
10 Column{
11     Button(onClick = {
12         launcher.launch(arrayOf("*/*"))
13     }){
14         Text(text="Select doc")
15     }
16     result.value?.forEach(){uri ->
17
18         uri?.let{
19             val file = saveUriToFile(context, it)
20             if(uri?.let { checkFileType(context, it) } != null &&
21                 checkFileType(context, uri) == "application/octet-stream" ){
22                 dataMap["bin"] = file?.let { file_ -> readShort(file_) }!!
23             }else{
24                 if (file != null && file.canRead() && file.readText().
25                     contains("start")) {
26                     dataMap["meta"] = file.readText().split(";")
27                 }else{
28                     if (file != null) {
29                         dataMap["bpm"] = file.readText().split(";")}
30                 }
31             }
32         }
33     }
34 }
35 }
36 }
37 return dataMap
38 }
39 }

```

Code 18: Code for filechooser functionality

## 5. SUMMARY

This thesis examined the design and implementation of a front-end application for real-time and long-term monitoring of electrocardiographic (ECG) data. The focus was on the

processing of ECG-data, Bluetooth Low Energy (BLE) communication and the visualization of ECG curves in a mobile application. For this purpose, an application was developed in Jetpack Compose that continuously receives and processes ECG data and displays it in a user-friendly way. A key challenge was the efficient processing of data streams that deliver new values every 8 milliseconds. This required optimized algorithms for storing, filtering and displaying the signals to ensure a smooth and delay-free visualization. In addition, mechanisms for recognizing response packets and reassembling fragmented data were implemented to ensure stable and reliable communication with the microcontroller. The developed application enables a reliable and intuitive visualization of ECG data in real time and thus provides a basis for further analysis and medical applications. s

## 6. RESULT

The development and implementation of the ECG monitoring application using Bluetooth Low Energy (BLE) yielded significant results in multiple aspects, including real-time data acquisition, and signal processing. The results of this work demonstrate the feasibility of using a mobile application for continuous and long-term ECG data recording while ensuring low power consumption and maintaining a stable BLE connection.

### 6.1. Successful Real-Time ECG-Data Acquisition

One of the core objectives of this thesis was to achieve reliable and continuous real-time ECG data acquisition from an external sensor via BLE. The final implementation successfully established a stable BLE connection that allows the application to receive ECG data at a sampling rate of 125 Hz.

The application was tested with simulated ECG signals, and later with real ECG data from the microcontroller. The results confirmed that the received data is transmitted with minimal latency, ensuring that real-time monitoring is not disrupted by delays or data loss. The packet structure and reassembly mechanisms implemented in the application effectively handled BLE's limited packet size, ensuring that all transmitted ECG data was correctly reconstructed.

### 6.2. Efficient Signal Processing and Visualization

The GraphEcgWidget component successfully visualized the ECG waveform in a smooth, high-resolution format. A major challenge in signal rendering was handling rapid incoming data updates without performance degradation. This was solved using Jetpack Compose's

Canvas API, which enabled optimized redrawing of only updated parts of the graph, ensuring high efficiency.

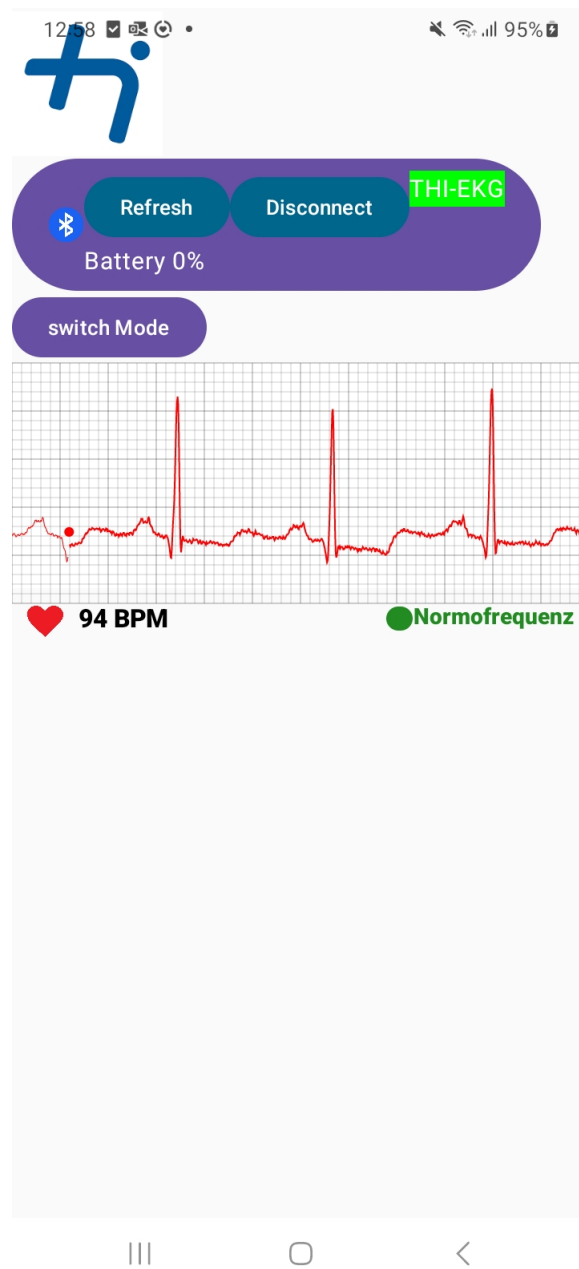
Further testing of data mapping accuracy showed that the ECG waveforms were correctly aligned and scaled to fit various screen sizes. The adaptive coordinate transformation algorithm ensured that signal peaks remained correctly proportioned regardless of device resolution, preventing distortions that could mislead users during analysis.

### **6.3. 24-Hour ECG Recording Mode**

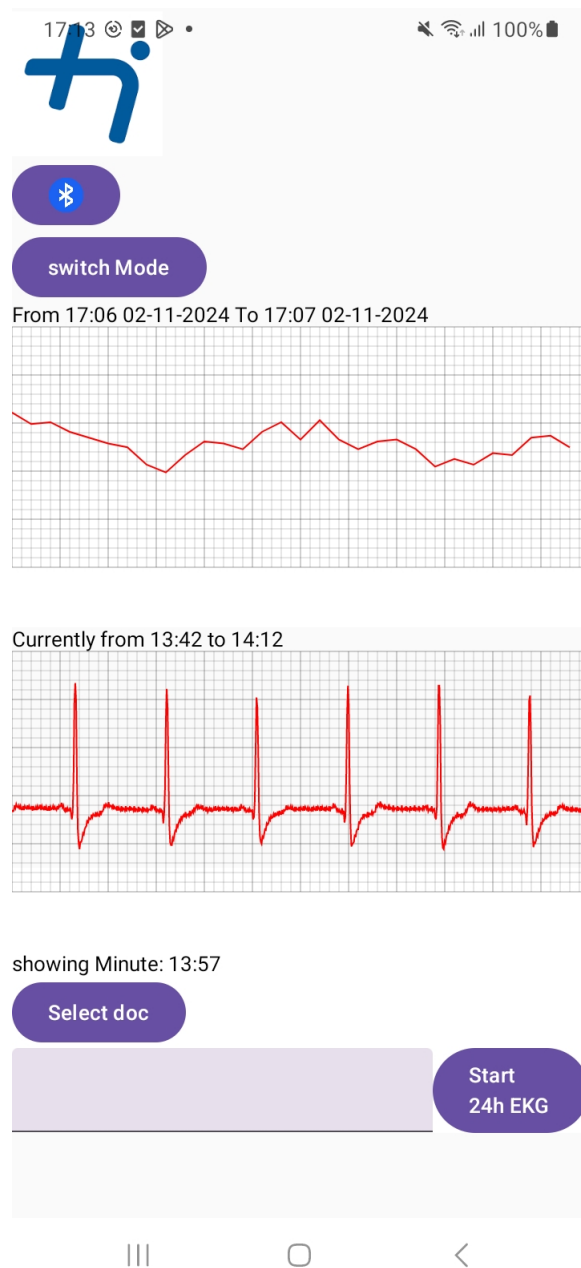
A key feature of the application was the implementation of a 24-hour recording mode, which allows long-term ECG data storage and retrieval. This feature was tested under continuous data acquisition conditions for multiple hours, simulating a full-day recording scenario.

The file storage system successfully handled large volumes of ECG data while maintaining fast retrieval times. By implementing a structured file format, the recorded ECG data could be efficiently loaded and visualized.

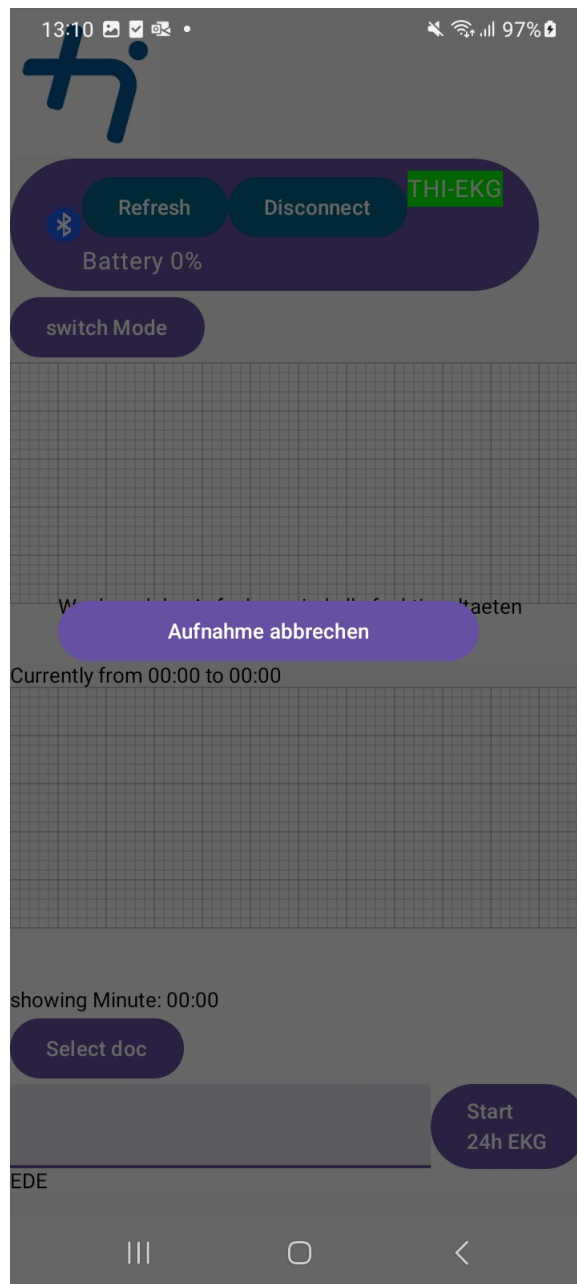
## 6.4. User Interface



**Figure 9:** Drawing ECG-Data in realtime



**Figure 10:** Displaying recorded ECG-Data and BPM (VitalMonitorWidget)



**Figure 11:** Recording Mode ("Stop Recording")

## 7. OUTLOOK

The current implementation of the frontend application for real-time and 24-hour ECG monitoring provides a substantial foundation for visualizing and analyzing electrophysiological data. However, there remain several opportunities for improvement and further development that can enhance the application's usability, efficiency, and functionality. One key area for improvement is the user interface and user experience. A more intuitive and customizable UI can enhance user interaction, making it easier for both medical

professionals and patients to navigate the application. Features such as customizable dashboards, improved accessibility, and better support for different screen sizes can significantly improve usability. Another important area is the integration of artificial intelligence to assist in anomaly detection. Machine learning models can be trained to recognize abnormal ECG patterns and provide early warnings about potential cardiac conditions. Implementing AI-based real-time classification and predictive analytics can enhance the system's capabilities and assist healthcare professionals in decision-making. The accuracy and efficiency of data processing can also be improved. Advanced filtering techniques can help reduce noise in the ECG signals, leading to more accurate readings. Optimizing data rendering and visualization can make the real-time display more fluid, especially when handling long-duration recordings. By addressing these areas, the application can evolve into a more robust and versatile tool, offering enhanced capabilities for medical professionals and patients alike. Future research should focus on improving usability, leveraging AI for anomaly detection, optimizing BLE communication, and ensuring compliance with medical standards.



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## A. APPENDIX

### A.1. Versions

This section presents fundamental versions of the required development resources used for this work.

The tests for this application during the development process, were done on a phone using the **Android version 13** which implies the usage of **API level 33**. Utilizing lower Android version than 12 (API Level 31,32) is not possible due to new runtime permissions added in Android 13, making it impossible to start the app<sup>7</sup>. Higher Android versions were not reviewed. More versions of the used resources, are presented in table 3.

**Table 3:** Versions of used resources

| <b>Ressource</b>                  | <b>Version used</b> |
|-----------------------------------|---------------------|
| Android version                   | 13                  |
| API Level                         | 33                  |
| Kotlin compiler extension version | 5.15.1              |
| minSDK                            | 30                  |

---

<sup>7</sup><https://developer.android.com/develop/connectivity/bluetooth/bt-permissions>